

Software Architecture - Quality Attributes

Time: 15 minutes.

* Indicates required question

1. Full Name *

2. Student ID *

3. Functionality ____.*

1 point

Check all that apply.

- ☐ is independent of any particular structure.
- ☐ is achieved by assigning responsibilities to architectural elements.
- ☐ does not interact with or constrain other qualities.

4. Stakeholders in the architecture include ____.*

1 point

Check all that apply.

- ☐ management.
- ☐ customers.
- ☐ developers.
- ☐ end users.
- ☐ marketing users.
- ☐ operators.
- ☐ maintainers.

5. Which requirement below is a quality goal? *

1 point

Check all that apply.

- ☐ The system shall be robust.
- ☐ The system shall be highly modifiable.
- ☐ The system shall be secure from unauthorized break-in.
- ☐ The product shall conserve battery life.

6. The parts of a quality attribute scenario include ____.

1 point

Check all that apply.

- ☐ Goal.
- ☐ Actor.
- ☐ Source of stimulus.
- ☐ Stimulus.
- ☐ Environment.
- ☐ Artifact.
- ☐ Response.
- ☐ Response measure.

7. ____ is a condition that will cause the system to respond.

1 point

Mark only one oval.

- ☐ Request
- ☐ Stimulus
- ☐ Error
- ☐ Perturbation

8. ____ is the part of the system affected by the stimulus. *

1 point

Mark only one oval.

- ☐ Context
- ☐ Process
- ☐ Artifact
- ☐ Component

9. The response measures of a performance scenario include ____.

1 point

Check all that apply.

- ☐ latency.
- ☐ throughput.
- ☐ downtime.

10. The response measures of an modifiability scenario include ____.

1 point

Check all that apply.

- ☐ the number of affected components.
- ☐ the amount of effort involved in modification.
- ☐ the maintenance downtime.

11. The tactics for a performance scenario include *

1 point

Check all that apply.

- ☐ Manage sampling rate.
- ☐ Limit event response.
- ☐ Prioritize events.
- ☐ Increase resources.
- ☐ Introduce concurrency.
- ☐ Maintain multiple copies of computations.

12. How to promote conceptual integrity? *

1 point

Check all that apply.

- ☐ Do regular code reviews with the development team.
- ☐ Establish conventions, such as structural rules or naming conventions.
- ☐ Split the development team into subteams for each component of the architecture.
- ☐ Provide document templates.

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